Dongho Kang

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RESEARCH INTERESTS

My research focuses on developing embodied intelligence for humanoid robots, enabling natural, effective, and adaptive whole-body loco-manipulation in real-world environments at human cadence.

PROFESSIONAL AFFILIATIONS & ACTIVITIES

RAI Institute, Zurich, Switzerland

- Research Scientist — Jun 2025 – Present
  - Led research on data-driven control frameworks to achieve for natural and steerable humanoid whole-body behaviors.
  - Developed robust perceptive locomotion control pipelines for legged systems.

NVIDIA, Zurich, Switzerland

- Deep Learning Intern — Jun 2018 – Dec 2018
  - Explored and developed deep learning algorithms for super-resolution and anti-aliasing within the DLSS pipeline.

CNP Technology Inc., Seoul, South Korea

- Hardware Engineer — Dec 2011 – Mar 2014
  - CAD-designed enclosures and managed end-to-end manufacturing processes for industrial and embedded PCs.

EDUCATION

ETH Zürich, Zurich, Switzerland

- Doctor of Science in Computer Science — Apr 2020 – Aug 2025
  - Main advisor: Prof. Dr. Stelian Coros
  - Second advisor: Prof. Dr. Marco Hutter
  - Topic: motion data-driven legged locomotion control via reinforcement learning and optimal control.
- Master of Science in Mechanical Engineering — Sep 2016 – Aug 2019
  - Advisor: Prof. Dr. Marco Hutter
  - Topic: physics simulation for legged robotics applications and reinforcement learning-based navigation

Seoul National University, Seoul, South Korea

- Bachelor of Science in Mechanical Engineering & Computer Science — Mar 2009 – Aug 2016
  - Advisor: Prof. Dr. Dongjun Lee

PUBLICATIONS AND PATENTS

JOURNALS

[1] Mike Zhang, Dongho Kang, Kevin Bergamin (co-first authors); advised by Alfred Rizzi, Jessica Hodgins, Scott Kuindersma, Yeuhi Abe, Sylvain Bertrand, and Farbod Farshidian, "HuMBLE: Human Motion-Driven Behavior Learning for Embodied Locomotion," 2026 (under review.)

[2] Taein Kwon, Ergys Ristani, Dongho Kang, Marc Pollefeys, and Li Guan, "Glove-based Microgesture Recognition Using Egocentric Vision and IMU Fusion," in *IEEE Robotics and Automation Letters (RA-L)*, 2026 (accepted.)

[3] Matthias Heyrman, Chenhao Li, Victor Klemm, Dongho Kang, Stelian Coros, and Marco Hutter, "Multi-Domain Motion Embedding: Expressive Real-Time Mimicry for Legged Robots," in *The International Journal of Robotics Research (IJRR)*, 2026 (accepted.)

[4] Lukas Molnar, Jin Cheng, Gabriele Fadini, Dongho Kang, Fatemeh Zargarbashi, and Stelian Coros, "Whole-body Inverse Dynamics MPC for Legged Loco-manipulation," in *IEEE Robotics and Automation Letters (RA-L)*, Nov 2026.

[5] Jin Cheng, Dongho Kang, Gabriele Fadini, Guanya Shi, and Stelian Coros, "RAMBO: RL-augmented Model-based Optimal Control for Whole-body Loco-manipulation," in *IEEE Robotics and Automation Letters (RA-L)*, Sep 2025.

[6] Taerim Yoon, Dongho Kang, Seungmin Kim, Jin Cheng, Minsung Ahn, Stelian Coros, and Sungjoon Choi, "Spatio-Temporal Motion Retargeting," in *IEEE Transactions on Robotics (T-RO)*, Aug 2025.

[7] Dongho Kang, Jin Cheng, Miguel Zamora, Fatemeh Zargarbashi, and Stelian Coros, "RL + Model-based Control: Using On-demand Optimal Control to Learn Versatile Legged Locomotion," in *IEEE Robotics and Automation Letters (RA-L)*, Oct 2023.

CONFERENCES

[1] Taerim Yoon, Dongho Kang, Kisang Park, Junha Cha, Stelian Coros, Sungjoon Choi, "Learning Unions of Convex Sets via Invertible Latent Decomposition for Path Planning," in *Conference on Robot Learning (CoRL)*, 2026.

[2] Taerim Yoon, Dongho Kang, Jin Cheng, Fatemeh Zargarbashi, Yijiang Huang, Minsung Ahn, Stelian Coros, and Sungjoon Choi, "Teaching Robots Like Dogs: Learning Agile Navigation from Luring, Gesture, and Speech," in *Conference on Robot Learning (CoRL)*, 2026.

[3] Yarden As, Chengrui Qu, Benjamin Unger, Dongho Kang, Max van der Hart, Laixi Shi, Stelian Coros, Adam Wierman, Andreas Krause, "SPiDR: A Simple Approach for Zero-Shot Safety in Sim-to-Real Transfer," in *Neural Information Processing Systems (NeurIPS)*, 2025.

[4] Fatemeh Zargarbashi, Jin Cheng, Dongho Kang, Robert Sumner, and Stelian Coros, "RobotKeyframing: Learning Locomotion with High-Level Objectives via Mixture of Dense and Sparse Rewards," in *Conference on Robot Learning (CoRL)*, 2024.

[5] Adrian Hartmann, Dongho Kang, Fatemeh Zargarbashi, Miguel Angel Zamora Mora, and Stelian Coros, "Deep Compliant Control for Legged Robots," in *International Conference on Robotics and Automation (ICRA)*, 2024.

[6] Daniel Widmer, Dongho Kang (co-first authors), Bhavya Sukhija, Jonas Hübotter, Andreas Krause, and Stelian Coros, "Tuning Legged Locomotion Controllers via Safe Bayesian Optimization," in *Conference on Robot Learning (CoRL)*, 2023.

[7] Dongho Kang, Flavio De Vincenti, Naomi C. Adam, and Stelian Coros, "Animal Motions on Legged Robots Using Nonlinear Model Predictive Control," in *International Conference on Intelligent Robots and Systems (IROS)*, 2022.

[8] Dongho Kang, Simon Zimmermann, and Stelian Coros, "Animal Gaits on Quadrupedal Robots using Motion Matching and Model-Based Control," in *International Conference on Intelligent Robots and Systems (IROS)*, 2021.

[9] Flavio De Vincenti, Dongho Kang, and Stelian Coros, "Control-Aware Design Optimization for Bio-Inspired Quadruped Robots," in *International Conference on Intelligent Robots and Systems (IROS)*, 2021.

[10] Changu Kim, Hyunsoo Yang, Dongho Kang and Dongjun Lee, "2-D Cooperative Localization with Omni-Directional Mobile Robots," in *International Conference on Ubiquitous Robots and Ambient Intelligence*, 2015.

THESIS

[1] Dongho Kang, "Animal Motion Imitation For Adaptive and Lifelike Control of Legged Robots," Doctoral thesis, Department of Computer Science, ETH Zürich, 2025.

[2] Dongho Kang, "End-to-End Collision Avoidance from Depth Input with Memory-based Deep RL," Master's thesis, Department of Mechanical and Process Engineering, ETH Zürich, Aug 2019.

PATENTS

[1] Fatemeh Zargarbashi, Robert Walker Sumner, Jin Cheng, Dong Ho Kang, Stelian Coros, "Methods and systems for robot keyframing and learning locomotion with high-level objectives," US Patent, 2024 (under review.)

INVITED TALK

- Institute of Robotics and Mechatronics, German Aerospace Center (DLR) — Jul 2026
  Oberpfaffenhofen-Weßling, Germany
- Illinois Robotics Group, University of Illinois at Urbana-Champaign — Mar 2026
  Champagne-Urbana, Illinois, United States (online)
- Laboratory for Intelligent Decision and Autonomous Robots, Georgia Tech — Feb 2026
  Atlanta, Georgia, United States
- Department of Artificial Intelligence, Korea University — Jan 2026
  Seoul, South Korea
- Department of Computer Science and Engineering, Seoul National University — Jan 2026
  Seoul, South Korea

- Machine Perception for Human Understanding, AI+X Summit 2025 — Oct 2025
  Zurich, Switzerland
- Department of Mechanical Engineering, Seoul National University — Oct 2025
  Seoul, South Korea (online)
- Artificial Intelligence Graduate School, UNIST — Oct 2025
  Ulsan, South Korea (online)
- Biomimetic Robotics Lab, Massachusetts Institute of Technology — Aug 2024
  Cambridge, Massachusetts, United States (online)
- Johou Systems Kougaku Laboratory, University of Tokyo — May 2024
  Tokyo, Japan
- Suzumori Laboratory, Tokyo Institute of Technology — May 2024
  Tokyo, Japan

TEACHING EXPERIENCE

ETH Zürich, Zurich, Switzerland

- Teaching Assistant, Stochastics and ML (A. Streich, C. Cotrini, F. Friedrich) — Spring 2025
- Teaching Assistant, Introduction to Machine Learning (F. Perez-Cruz, F. Yang) — 2024
- Teaching Assistant, Computer Science (M. Fischer, F. Friedrich) — 2023
- Teaching Assistant, Digital Humans (S. Coros, Siyu Tang) — 2023
- Teaching Assistant, Linear Algebra (Ö. Imamoglu, O. Sorkine-Hornung) — 2022
- Teaching Assistant, Computational Models of Motion (S. Coros, B. Thomaszewski) — 2021 – 2022
- Teaching Assistant, Visual Computing (S. Coros, M. Pollefeys) — 2020 – 2021

Seoul National University, Seoul, South Korea

- Mentor, SNU Samsung Convergence Software Course Program — 2015
- Teaching Assistant, MAE 446.204A: Dynamics — 2014
- Teaching Assistant, PA 034.013: Basic Physics 2 — 2011

TECHNICAL SKILLS

Programming and Software

C/C++, Python, Matlab, Unix/Linux, Pytorch, ROS, Open Dynamics Engine, IsaacSim, MuJoCo

Experience with Robots

UnitreeRobotics AlienGo/A1/Go1/Go2/B2/G1, ANYbotics ANYmal, Boston Dynamics Atlas

SERVICES

Reviewer

T-RO, T-RL, TIE, RA-L, ICRA, IROS, RSS, CoRL, Humanoids, BioRob, Eurographics

LANGUAGES

- Korean: Native language.
- English: Fluent.

REFERENCES

- **Prof. Dr. Stelian Coros**
  Associate Professor, ETH Zürich
  scoros@inf.ethz.ch
- **Prof. Dr. Marco Hutter**
  Professor, ETH Zürich / Senior Director, RAI Institute
  mahutter@ethz.ch
- **Prof. Dr. Dongjun Lee**
  Professor, Seoul National University
  djlee@snu.ac.kr